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Question 2C
Create a behavioral Verilog description for the FSM using the template provided.
Assuming clock-synchronous reset on high reset input.

syntax references used:
https://chipverify.com/verilog/verilog-always-block
https://coddy.tech/learn/verilog/fundamentals/concatenation_operator for sanity check
https://chipverify.com/verilog/verilog-case-statement
==========

```
module state_machine (clk, reset, taken, predict);

        input clk, reset, taken;
        output predict;

        reg [1:0] cs; //use bit width and number of variables as desired
        //using the register cs to store states 0 to 3.


        //Fill in….
        always @(posedge clk) begin
                if(reset) begin
                        cs = 2'b00; //reset state to 00.
                end else begin
                        case ({taken, cs}) begin //3 to 2 mux
                                3'b000: begin
                                        cs = 2'b01;
                                        predict = 1'b1;
                                end
                                3'b001: begin
                                        cs = 2'b11;
                                        predict = 1'b1;
                                end
                                3'b010, 3'b011: begin
                                        cs = 2'b11;
                                        predict = 1'b0;
                                end
                                3'b100, 3'b101: begin
                                        cs = 2'b00;
                                        predict = 1'b1;
                                end
                                3'b110: begin
                                        cs = 2'b00;
```

```
                predict = 1'b0;
            end
            3'b111: begin
                cs = 2'b10;
                predict = 1'b0;
            end
        endcase
    else
end
endmodule
```